INCH-POUND

MIL-P-70456C (AR)
AMENDMENT 3
5 March 1992

SUPERSEDING
AMENDMENT 2
28 May 1991

MILITARY SPECIFICATION

PROPELLANT, STICK, M31A1E1
FOR USE IN CHARGE PROPELLING M203A1

This Amendment forms a part of Military Specification MIL-P-70456C (AR) dated 28 April 1989, and is approved for use within the US Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of Department of Defense.

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3.9.1.2 Delete in its entirety and substitute new paragraph 3.9.1.2:

"3.9.1.2 145 degrees F pressure. At standard weapon conditions and recommended charge weight, the average pressure at 145 degrees F shall not exceed 56,000 pounds per square inch (psi) (see 4.4.2.5.1)."

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Add the following paragraphs:

"4.1.3 Nonconforming material. The contractor's quality program plan (see 6.2) shall include an effective and positive system for controlling nonconforming material to include mixes, batches, or sublots of propellant that do not meet compositional requirement (see 3.2 and 4.1.3.1). The system shall include procedures for nonconforming material identification, segregation, and disposition. All nonconforming supplies shall be positively identified to prevent unauthorized use, shipment and intermingling with conforming supplies.

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4.1.3.1 Manufacturing process. The contractor's manufacturing process shall be documented as part of the quality program plan (see 6.2). Under a batch manufacturing process, the contractor shall maintain an effective and positive system (weighing each ingredient) for controlling amounts of ingredients charged to each mixer to assure the material meets the compositional requirements (see 3.2). Under a continuous process where the batch system is not applicable, the contractor shall maintain an effective and positive system for controlling amounts of ingredients charged to each mixer to assure the material meets the compositional requirements (see 3.2). For an unproven propellant manufacturing process which is significantly different from the present propellant manufacturing process, the contractor shall submit his proposed process control system for meeting compositional requirements in the form of a process control document to the contracting officer for approval by the technical agency. The contractor shall not initiate production without written approval of the process control document from the contracting officer. Once approved the contractor shall not change the process without approval from the contracting officer (see 6.13)."

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4.3.2.2 Delete in its entirety and substitute new paragraph 4.3.2.2:

"4.3.2.2 Testing to be performed. The following test program shall be used for the first article test. (The first article test is a qualification test for a new contractor or a major process change. Once a contractor has demonstrated the ability to produce acceptable stick propellants the first article test may be waived by the Contracting Officer. The following test program (see Table III) is the minimum test requirement for the first article test and shall be used when the stick propellant manufacturing process is not significantly different from the present propellant manufacturer. If the new contractor's process or the proposed process change is found to be significantly different from the present manufacturing process, additional testing may be required to qualify the process difference.)

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TABLE III

First Article Test

Test Phase	Temp (Deg F)	No. of Calibration Rds	No. of Test Rds	TECOM TOP
Propellant Checkout	70		10	TOP-4-2-504
	145		10	TOP-4-2-504
	-60		10	TOP-4-2-504
Sequential Environ-mental	145	15	60	TOP-4-2-504
	-60	15	60	TOP-4-2-504"

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4.4.2.5 Delete in its entirety and substitute new paragraph 4.4.2.5:

"4.4.2.5 Permissible individual maximum pressure. If at any time during testing, an individual pressure greater than that specified in 3.9.1.1 is observed, testing shall cease and the lot shall be rejected."

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4.4.2.5.1 Delete in its entirety and substitute new paragraph 4.4.2.5.1:

"4.4.2.5.1 Average 145 degrees F pressure. If the average pressure at 145 degrees F exceeds 56,000 pounds per square inch (psi), the lot shall be referred."

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4.5.1.3 Delete in its entirety and substitute new paragraph 4.5.1.3:

"4.5.1.3 Length. Five hundred (500) sticks shall be randomly selected from each sublot and measured by gage for length. If one stick is found to be exceeding the maximum length requirement, the sublot shall be rejected. Sticks with a length shorter than the minimum requirement shall be classified by the degree of deviation from the minimum length.

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The basis for rejection of a sublot shall be: (1) Reject if more than 14 sticks are found shorter than 28-13/16 inches. (2) Reject if more than 7 sticks are found shorter than 28-9/16 inches. (3) Reject if one or more sticks are found shorter than 28-1/2 inches. A 100% inspection on stick length shall be done on the rejected sublot. Each stick shall be measured by gage and any stick which does not meet the length requirement should be removed from the sublot. The final blending on the entire lot should not be initiated until all sublots are found to be meeting the stick length requirement. The gage design for propellant length measurement shall be submitted to the technical agency for approval."

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* 4.5.1.4.1 Delete in its entirely and substitute new paragraph 4.5.1.4.1:

"4.5.1.4.1 Carbon black. The actual percentage of carbon black shall be determined in accordance with Method 308.1 or 309.1, MIL-STD-286."

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4.5.3.1.1 Add the following to the end of the para 4.5.3.1.1:

"The manufacturer of the combustible cases used in the calibration components for the propellant acceptance test shall be specified in the Propellant Acceptance Sheet along with the recommended charge weight (see 6.9.3)."

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6.2 Delete in its entirety and substitute new paragraph 6.2:

"6.2 Ordering data.

6.2.1 Contractor's quality program. Contracts should identify MIL-Q-9858 as the applicable higher level contract quality control requirement and specify data item description DI-R-5297A.

6.2.2 Data requirements. One copy of the Propellant Description Sheet and one copy of all ballistic test data for each lot of propellant shall be forwarded to ARDEC, Picatinny Arsenal, NJ 07806-5000, ATTN: SMCAR-QAR-R, and SMCAR-AEE-BP. Individual measurements of the total volatiles, stick weights, stick bundle weights and dimensions shall be recorded in the Propellant Description Sheet for proper evaluation of stick propellant."

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Add the following new paragraph:

"6.9.3 Recommended charge weight. The recommended charge weight as specified in the Propellant Acceptance Sheet may need final weight adjustment if the manufacturer of the combustible case components used for the propelling charge loading, assembling and packing is different from the manufacturer of the combustible case components used for calibration components which are used for the propellant acceptance test. The weight adjustment for the recommended charge weight shall be obtained from the first article test for the new contractor for manufacturing of the combustible case for the propelling charge as specified in MIL-C-70461."

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